D02 · Supply Planning & Scheduling

04 · Supplier capacity management & supply commitment

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1 WHAT IT IS

Supplier capacity management and supply commitment build the supplier's real, finite capacity into the plan — through capacity collaboration, reservation, and flexibility commitments — so the plan promises only what the supply base can actually deliver, instead of exploding demand against an implicit assumption of infinite upstream capacity and discovering the shortfall when material fails to arrive.

2 WHY IT MATTERS

Tsay 1999 **(Tsay, 1999 — peer-reviewed)** analyzed the quantity-flexibility contract and showed that explicit commitments — a baseline volume the buyer guarantees and a flexibility band the supplier agrees to cover — align incentives and let both parties carry less buffer than they would acting independently. A plan that treats supplier capacity as infinite carries no such commitment: it assumes any quantity can be sourced at the planned lead time, which is false the moment demand spikes or a supplier is shared across customers. The failure surfaces late and expensively — **a material shortfall discovered at the planned receipt date leaves no recovery time**, converting a capacity conversation that could have happened months earlier into an expedite, a line-down, or a missed customer commitment.

3 HOW IT IS DONE

1. **Model supplier capacity as a finite constraint.** Capture each critical supplier's real capacity, not an assumed-infinite default, and plan against it. A supply plan that ignores the upstream ceiling is a forecast of receipts that cannot happen.
2. **Share the forward demand signal with suppliers.** Give critical suppliers visibility to the planning horizon, not just firm POs. A supplier that sees demand coming can position capacity; a supplier that learns of it at order release can only react.
3. **Establish commitment and flexibility agreements.** Define the baseline volume the buyer commits to and the upside flexibility band the supplier agrees to cover, with the notice required. This converts an implicit assumption into an explicit, enforceable capability — the core of the quantity-flexibility logic.
4. **Account for shared capacity across customers.** A supplier's capacity is rarely dedicated; model the share realistically available, especially in allocation conditions. Planning against a supplier's total capacity when you hold a fraction of it is how a confident plan meets an empty dock.
5. **Surface capacity gaps early, in the S&OP supply review.** A supplier capacity shortfall is a supply-review escalation, resolved months ahead through dual-sourcing, pre-builds, or demand shaping — not a surprise at the receipt date.
6. **Monitor supplier delivery against commitment.** Track on-time-in-full and capacity adherence per supplier as a standing metric. A supplier consistently missing its committed band is a capacity risk to design around now, not a recurring expedite to absorb forever.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> An OEM planned against assumed-infinite supplier capacity on a key constrained component — **baseline: shortfalls discovered at receipt date, ~6 expedite events per quarter, two line-down incidents in a year, no forward capacity visibility**. It modeled supplier capacity as finite, shared the planning horizon, and signed baseline-plus-flexibility commitments on critical parts. Results: **capacity gaps surfaced 8–12 weeks ahead in the supply review, expedite events down ~60%, line-downs eliminated** — the shortfall conversation simply moved months upstream of the dock.

IN PRACTICE

Forward visibility to critical suppliers

Share the planning-horizon demand signal with critical suppliers, not just firm orders, so they can position capacity ahead of need. Visibility is the cheapest form of capacity assurance — it costs a data feed and buys the supplier the lead time to prepare.

Baseline-plus-flexibility commitments

Negotiate explicit commitments on critical parts: a guaranteed baseline volume and a flexibility band with defined notice. This is the contractual form of the quantity-flexibility logic and the mechanism that lets both sides plan with confidence instead of mutual hedging.

Capacity gaps in the supply review

Route supplier-capacity shortfalls into the S&OP supply review as standing escalations, resolved through dual-sourcing, pre-builds, or demand shaping months ahead. The supply review exists precisely to confront the plan with finite capacity — supplier capacity belongs there, not at the dock.

Supplier capacity-adherence metrics

Track each critical supplier's delivery against its committed band and capacity. Persistent under-delivery is a structural risk to design around — through dual sourcing or buffer placement — not an operational nuisance to absorb cycle after cycle.

WHAT YOU WOULD USE

SUPPLIER CAPACITY / SUPPLY COLLABORATION

SAP IBP Response & Supply / o9 / Kinaxis — enterprise tier-1 — Finite-capacity supply planning with supplier collaboration portals, capacity commitments, and forward visibility built into the plan. **(SAP, o9, Kinaxis — vendor)** The choice when supplier capacity must be a modeled constraint inside the integrated plan.

SAP Ariba / supplier-collaboration networks — Supplier-facing collaboration and commitment networks that share forecasts and capture capacity commitments. **(SAP Ariba, supply-network vendors — vendor)** A strong fit where the priority is the supplier-collaboration layer over the planning math.

Multi-tier supply-visibility specialists — Dedicated tools for multi-tier supplier capacity and component visibility, common in electronics and automotive. **(specialized vendors — vendor)** The vertical option when constrained components and deep multi-tier exposure are the core risk.

For operations that don't need a platform — A shared rolling forecast to critical suppliers, written baseline-plus-flexibility commitments, and supplier capacity tracked as a finite line in the supply-review workbook. The commitment and the forward visibility — not the portal — are what move the shortfall conversation upstream of the dock.

4 SO WHAT

Operations that model finite supplier capacity and secure it with commitments surface shortfalls months ahead, while there is still time to act; operations that assume infinite upstream capacity discover them at the dock, when the only options left are expensive. The infinite-capacity assumption is the most comfortable lie in supply planning — it makes every plan look feasible right up to the receipt date it isn't. Commitment is what turns supplier capacity from a hope into a number.

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